EmbodyHOI: Embodiment-Aware Generation of Hand–Object Interactions for Humanoid Dexterous Manipulation

Anonymous Authors

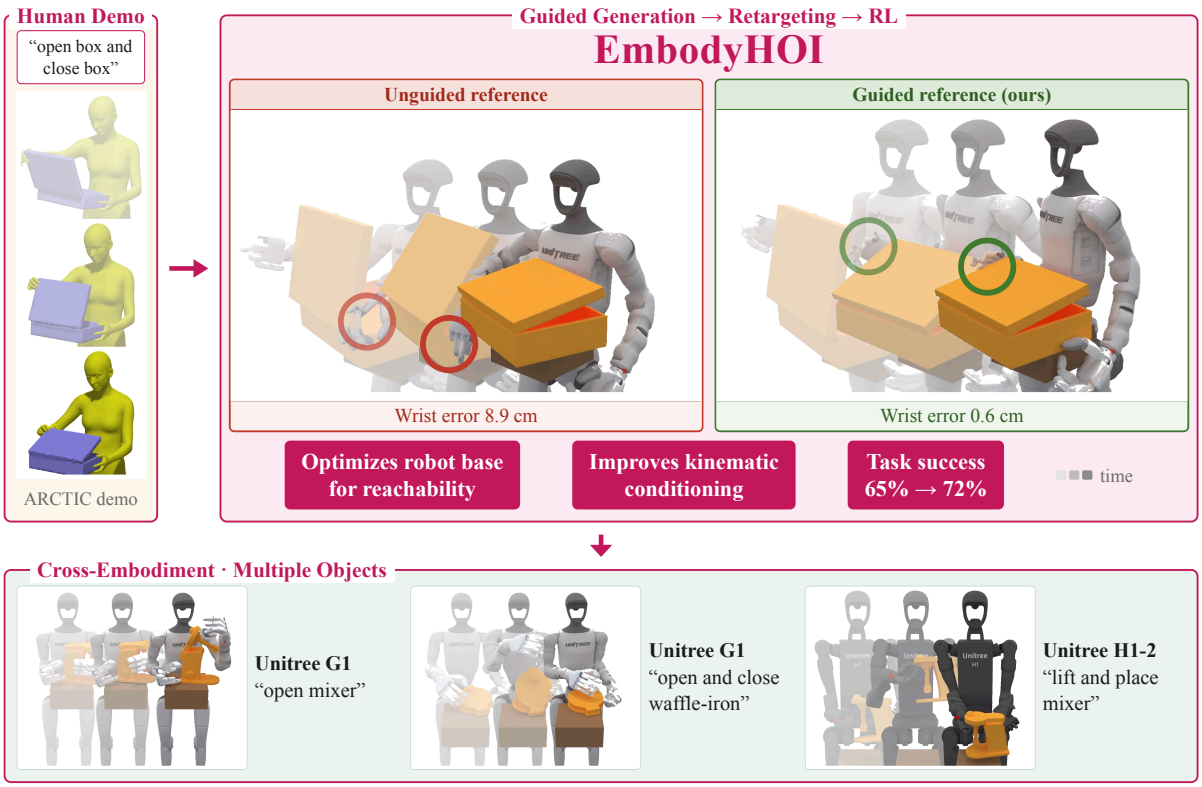


Fig. 1: EMBODYHOI steers a human-space HOI generator toward motions a specific humanoid can reach. During sampling, a differentiable IK solve on the target robot scores the wrist trajectories, its gradient is fed back into the sampler, and the robot's base placement is optimized jointly.

***Abstract*— Humanoid dexterous manipulation needs reference motions that are both semantically rich and kinematically feasible for the target robot. Text-conditioned hand–object interaction (HOI) generators supply diverse, language-specified demonstrations, but they synthesize motion in human (MANO) space. The embodiment gap is therefore deferred to post-hoc retargeting, where wrist trajectories the robot cannot reach break contact and degrade downstream policy learning. Our key idea is that reachability should constrain the interaction *while it is being generated*, not after it is fixed. We present EMBODYHOI, which runs a differentiable inverse-kinematics (IK) solve inside the sampling loop of a frozen flow-matching HOI generator, feeding its gradient back into the sampler and jointly optimizing where the humanoid stands. Because guidance acts only at inference, one generator transfers across embodiments without retraining. On ARCTIC and two contrasting humanoids (Unitree G1 and H1-2), guidance cuts the retargeted wrist error from 8.9 cm to 0.6 cm while leaving text-to-HOI generation quality intact, and raises task success from 68% to 79% on 50 random tasks on G1 and from 62% to 76% on a four-task suite with the H1-2.**

[FIX] Doubled preposition ("on 50 random tasks on G1"), and the two halves are asymmetric — the first names a task count, the second a suite. Minor, but it is the second-last sentence of the abstract, where a stumble is expensive.

REPLACE: *raises task success from 68% to 79% on 50 random tasks on G1 and from 62% to 76% on a four-task suite with the H1-2.*

WITH: `raises task success from 68\% to 79\% on 50 random G1 tasks and from 62\% to 76\% on a four-task H1-2 suite.`

[NOTE:] *The abstract is much stronger than the previous draft* **— the explicit insight sentence ("reachability should constrain the interaction while it is being generated") and the promoted negative result are both doing exactly what the skill's Version 2 template asks, and swapping IK energy for the 8.9 cm→0.6 cm wrist error gives a reader something they can calibrate. Nothing else here needs changing. Critically, cheaper operators do not substitute: post-hoc wrist refinement and compute-matched best-of-N selection both reach *lower* IK energy than guidance yet train worse policies, because reachability alone is blind to whether the fingers and object still form a coherent interaction.**

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I. INTRODUCTION

Dexterous manipulation with multi-fingered hands remains the hard case for humanoids that already walk and loco-manipulate [1]–[4]. The robot must coordinate reaching,

finger contact, and whole-arm motion under embodiment-specific kinematic constraints. Small differences in finger morphology, joint limits, and wrist reachability can decide whether a human-like interaction is executable at all.

We study how to *generate* such interactions: given a language prompt and an articulated object, produce a bimanual hand–object trajectory that a specific humanoid can actually execute.

A complementary route to end-to-end policies is generate-then-track: a generator proposes diverse task-conditioned reference motions, and a physics-based RL policy learns to execute them under contact and actuation limits [5]. But motion that satisfies the text intent in human space can become unreachable, distorted, or contact-inconsistent once mapped to a robot: the generator has no term coupling the wrist to a robot's kinematic chain, so infeasibility is discovered only after finger and object motion are committed. Prior works either train the prior in robot motion space [6], which rules out cross-embodiment reuse and rarely models fine hand–object structure, or repair the motion after it is generated [7], [8], which inherits whatever the retargeted wrists cannot reach.

We take a different approach: keep a MANO [9] HOI generator as a *reusable* prior and make its samples embodiment-aware *during generation* (Fig. 1). Building on the text-conditioned flow-matching HOI model MAD-HOI [10], we introduce an IK guidance module that evaluates the decoded wrist trajectories on the target humanoid through a differentiable kinematics solver. We back-propagate the gradient of the resulting energy into the sampler, steering generation toward motions the robot can realize. Because MAD-HOI generates tokens over several unmasking steps, a wrist correction made in one step is visible to the finger and object tokens generated in the next, so the interaction re-forms around the corrected wrist. A second difficulty is specific to full humanoids: for floating hands or fixed-base arms the user can place the initial hand–object relationship manually, but for a humanoid the base placement decides whether the arms stay reachable over the whole interaction, and fixed-base IK can wrongly reject feasible motions. We therefore fold base-placement optimization into the IK objective. Finally, we retarget the guided motions and train a PPO policy on them with separate imitation rewards for arm and hand, because the two differ in error scale and contact requirements.

Reachability, it turns out, is not a quantity worth optimizing directly: post-hoc IK refinement of a finished sample and compute-matched best-of-N selection both reach lower IK energy than guidance yet train worse policies, because refinement moves the wrists after the fingers and object are fixed and selection returns a different interaction. Guidance costs $32\times$ a native sample (5 s to 160 s on an A100), paid once per reference.

Our contributions are as follows. *(1)* We run a differentiable IK solve inside the sampling loop of a frozen HOI generator, so the motions it produces are reachable by a specific humanoid — with no retraining, which is what lets one generator serve several robots. *(2)* We fold the robot's base placement into the same objective, which removes near-singular arm configurations (2.6% to 0.2% of samples) and, re-solved inside the loop, contributes 2.8 of the 5.7 points guidance adds. *(3)* We show that reachability is the wrong optimization target: lower IK energy from post-hoc refinement or best-of-N does not yield better policies, so the gain comes from correcting the requested sample. *(4)* Retargeting the guided references and training PPO with decoupled arm and hand rewards raises success from 68% to 79% on 50 random ARCTIC tasks (G1) and 62% to 76% on the H1-2, and the same objective lowers Text2HOI's IK energy without modification.

[FIX] The four contributions now run together as one prose block, because `\begin{compactitem}` is commented out at line 121 with no `\end`. Contributions are the most-scanned twelve lines in the paper and a reviewer skimming for them will not find a visual anchor. `compactitem` costs almost no vertical space over running text — roughly two lines for all four items — so this is a cheap recovery. If you need those two lines, keep the prose but bold the leads.

REPLACE: *`% \begin{compactitem}` (and the missing `\end{compactitem}`)*

WITH: `\begin{compactitem}` before item (1), `\item` at the head of each of the four, and `\end{compactitem}` after item (4). Failing that, open each with a bolded lead — `\textbf{IK guidance inside the sampling loop.}`, `\textbf{Base placement in the same objective.}`, `\textbf{Reachability is the wrong target.}`, `\textbf{Downstream and cross-embodiment gains.}` — which gives the skim-anchor at zero extra lines.

[RISK:] *The scope sentence has been dropped.* The previous draft ended the introduction with "Our scope is upper-body and kinematic: the lower body is fixed … all evaluation is in simulation". It is now only in Sec. V, eight pages later. That sentence was doing real work: it pre-empts the "unrealistic setting" objection (skill review dimension 5.1) at the moment a reviewer forms their first impression, and a reviewer who reaches the experiments still expecting whole-body real-robot results will feel misled rather than informed. Recommend restoring it verbatim as the last line of Sec. I: `Our scope is upper-body and kinematic: the lower body is fixed (the base is optimized, not walked), guidance reasons about reachability rather than dynamics or contact stability, and all evaluation is in simulation (Sec. \ref{sec:limitations}).`

II. Related Work

Human and hand–object motion synthesis. Text-to-motion generation for full-body SMPL [11] motion spans pose-space diffusion [12], [13] and masked-token generation [14]; test-time guidance adds spatial control over keyframes, trajectories, and contacts, all specified in human space [15], [16]. Full-body HOI methods [17], [18] model contact only coarsely at the body surface. Hand-centric methods parameterized by MANO [9] target fine-grained finger–object contact on datasets such as ARCTIC [19] and GRAB [20], from grasp synthesis [20], [21] to language-conditioned sequence generation [10], [22], [23]. However, none of these generators accounts for a robot embodiment; our guidance has no human-space target and instead scores the wrist against a robot's kinematic chain, so its optimum moves when the robot or its base changes.

Retargeting to bridge the embodiment gap. Human-to-robot retargeting is typically keypoint-based optimization

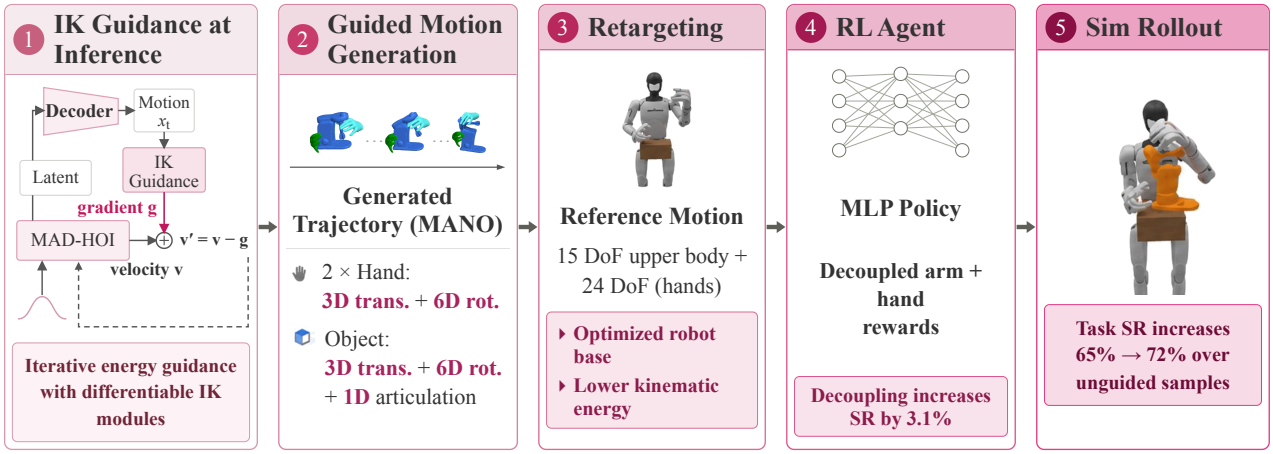


Fig. 2: EMBODYHOI makes a frozen HOI generator embodiment-aware at inference. During flow-matching sampling, a differentiable IK module on the target humanoid scores the decoded wrist trajectories, and we add its gradient to the learned velocity field (energy guidance). We then retarget the guided samples and train a PPO policy on them with decoupled arm and hand rewards; changing the robot changes only the IK module.

of joint angles and end-effector error [24], [25], allowing teleoperation [25] and robot-space datasets [6]. Dexterous-hand retargeting [26], [27] is a standard preprocessing step for RL from human demonstrations. ManipTrans [28] pretrains a hand-trajectory imitator before residual fine-tuning, DexMachina [29] uses retargeted joints and contacts as auxiliary rewards in a decaying virtual-object control curriculum, DexMan [30] extends this to RGB video, and HERMES [31] unifies heterogeneous motion sources for mobile bimanual manipulation. However, ManipTrans and DexMachina command a floating 6-DoF wrist without arm kinematics, and DexMan, which does drive a G1 arm through IK, fixes the robot at the origin and retargets a video-derived reference post hoc; in all of them the reference is fixed and arm reachability, our central concern, is never optimized. We reuse their object-side recipe (Sec. III-C) and treat the reference itself as the variable.

Generative priors for policy learning. Example-guided RL [32], [33] extends to HOI imitation [34], [35] and to tracking-as-reward execution [5], [36]. The two works closest to ours resolve the embodiment mismatch after the motion exists. Chen et al. [8] guide an RL finger controller with a generative prior that emits only wrists, fixed before the robot is considered, and DemoDiffusion [7] denoises a retargeted human trajectory with a robot diffusion policy that stays close to that trajectory and so inherits its reachability. DreamControl-v2 [6] avoids retargeting by training in robot space, and fine-tuning a human-space generator on reachable data would likewise remove our inference cost; both need robot-feasible data for every embodiment, which a method like ours would have to produce, and yield one generator per robot. However, none of them keeps one frozen human-space generator: we resolve reachability inside its sampling loop and handle base placement for a full humanoid arm chain.

III. METHOD

EMBODYHOI makes a *frozen* HOI generator embodiment-aware at inference time (Fig. 2): the generator needs only a decodable wrist pose (Sec. III-A), a differentiable IK energy on the target humanoid steers its sampler and solves for the robot's base (Sec. III-B), and we retarget the guided motion and train a PPO tracking policy (Sec. III-C).

A. Text-to-HOI Generation

We adopt MAD-HOI [10] as the generator (Text2HOI in Sec. IV-D). Given a prompt and an articulated object mesh, it produces an object trajectory and articulation $O = (T^t_{\text{obj}}, R^t_{\text{obj}}, \alpha^t_{\text{obj}})$ and MANO motion per-hand $H = (\boldsymbol{\tau}^t_h, \boldsymbol{\phi}^t_h)$ (wrist translation and 16 joint rotations in 6D form). Motion is represented in a decoupled latent space with separate object, left-hand, and right-hand tokens. A masked autoregressive model generates these tokens, and a flow-matching head denoises the masked ones. The head is a learned velocity field $\mathbf{v}_\theta(\mathbf{x}, t, \mathbf{z})$ that defines the probability-flow ODE $\dot{\mathbf{x}} = \mathbf{v}_\theta(\mathbf{x}, t, \mathbf{z})$, conditioned on the transformer context $\mathbf{z}$ from unmasked tokens and text. We use the ARCTIC dataset [19] (4.6k sequences, 11 articulated objects), whose bimanual articulated interactions are the hardest case for a humanoid arm chain.

B. Embodiment-Aware IK Guidance

Only the wrist component of H constrains the robot arm: $\boldsymbol{\tau}^t_h \in \mathbb{R}^3$ and $\boldsymbol{\phi}^t_h \in SO(3)$ for $h \in \{L, R\}$. Nothing ties these trajectories to the deployment robot, so per-frame IK retargeting is often ill-conditioned under large embodiment mismatch. The manipulator Jacobian of arm h with joint vector $\mathbf{q}^t$, $\mathbf{J}_h(\mathbf{q}^t) = \partial \operatorname{vec}(\mathbf{T}^t_{\text{ee},h})/\partial \mathbf{q}^t \in \mathbb{R}^{6\times n}$, loses rank near singular configurations. We monitor its minimum singular value $\sigma_{\min}(\mathbf{J}_h)$ and condition number $\kappa(\mathbf{J}_h)$ as proximity measures, and bias *generation* away from such configurations.

Energy guidance. Following classifier-style guidance for flow models [37], we add the gradient of an energy to the

velocity field during sampling,

$$\dot{\mathbf{x}} = \mathbf{v}_\theta(\mathbf{x}, t, \mathbf{z}) - w\,\nabla_{\mathbf{x}} E(\mathbf{x}), \tag{1}$$

with guidance weight w. The decoder $\mathcal{D}(\mathbf{x})$ maps the current latent to wrist trajectories, which serve as task-space IK targets on the robot.

Joint arm and base IK. At each timestep we optimize $\mathbf{u}^t = [\mathbf{q}^t; \mathbf{b}^t]$, arm joints and a base translation $\mathbf{b}^t \in \mathbb{R}^3$ (base orientation fixed). With world-frame wrist target $\mathbf{T}^t_{w,h}$ and base transform $\mathbf{T}^b$ built from $\mathbf{b}^t$, the pose residual is

$$\mathbf{e}^t_h = \log\left((\mathbf{T}^t_{w,h})^{-1}\mathbf{T}^b\,\mathrm{FK}_h(\mathbf{q}^t)\right) \in \mathbb{R}^6 \tag{2}$$

and additional residuals penalize base displacement from a prior $\mathbf{b}_0$ (root prior), rest-pose deviation, joint-limit violation, temporal smoothness, and violation of a clearance radius around the object. A secondary objective acts in the null space of the wrist task: it improves arm manipulability without moving the wrist. All weights are set once and shared by both embodiments and all tasks.

A fixed base makes reachability a property of the trajectory alone, so a feasible interaction performed from the wrong standing position is indistinguishable from an infeasible one, and fixed-base IK rejects both.

[FIX] Right sentence, wrong position. This is the *motivation* for putting $\mathbf{b}^t$ in the solve vector, but it now sits after the design is fully specified and before the LM solve — so it interrupts input→steps→output at the last step. The skill's module pattern is motivation, then design, then advantage. Move it up to open the **Joint arm and base IK.** paragraph, i.e. delete it here and paste it immediately after that bold heading, before "At each timestep we optimize …". One word changes so it reads as motivation rather than as an aside.

REPLACE: *A fixed base makes reachability a property of the trajectory alone, so a feasible interaction performed from the wrong standing position is indistinguishable from an infeasible one, and fixed-base IK rejects both.*

WITH:
```
We solve for the base because a fixed base
makes reachability a property of the trajectory
alone: a feasible interaction performed from the
wrong standing position is then indistinguishable
from an infeasible one, and fixed-base IK rejects
both.
```
— placed directly after **Joint arm and base IK.**

Stacking residuals into $\mathbf{r}(\mathbf{u})$ and solving with Levenberg–Marquardt (10 iterations) gives $\mathbf{u}^{t*}$, and the guidance energy is the mean squared residual over the horizon,

$$E(\mathbf{x}) = \tfrac{1}{T}\textstyle\sum_{t=1}^{T} \|\mathbf{r}(\mathbf{u}^{t*})\|_2^2. \tag{3}$$

Although the solve optimizes a base translation per timestep, the base is fixed during execution, so we average the per-timestep estimates into a single execution pose $\mathbf{b}_0 \leftarrow \mathrm{mean}_t(\mathbf{b}^{t*})$, which becomes the root prior for the next unmasking iteration. This single execution pose lets one generator serve robots of different proportions: a 45 cm G1 arm and a 68.5 cm H1-2 arm must stand in different places for the same interaction (Sec. IV-D).

Gradients through the solver. We differentiate implicitly through the LM solve in JAX, so $\partial E/\partial(\boldsymbol{\tau}, \boldsymbol{\phi})$ is well defined, and a JAX–PyTorch bridge propagates it through $\mathcal{D}$ to the latent $\mathbf{x}$. We mask guidance to hand tokens only, which preserves the generator's decoupled object–hand structure; guidance never edits object motion.

Why guidance must act during generation. MAD-HOI generates sequences by iterative unmasking/refinement; we inject guidance into the inner ODE of every unmasking step and refresh the base prior from the partially decoded trajectory (Alg. 1). Guidance thus edits the wrist *while* finger and object tokens are still being generated, so the interaction adapts to the new wrist, and it changes the sampling distribution rather than ranking its draws (Sec. IV-A).

Algorithm 1 One outer unmasking step with IK guidance

Require: partial latents $\mathbf{L}$, outer step s, IK module $\mathcal{G}$, weight w, base prior $\mathbf{b}_0$

sample mask $\mathcal{M}$ (cosine schedule); build $\mathbf{L}_{\mathrm{mask}}$
$\mathbf{z}_{\mathcal{M}} \leftarrow \text{MAD-HOI}(\mathbf{L}_{\mathrm{mask}}, \mathbf{c}_{\mathrm{text}})$; $\mathbf{x} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
for flow time $t = 0 \to 1$ (RK4, 20 steps) **do**
  $\mathbf{v} \leftarrow \mathrm{MLP}(\mathbf{x}, t, \mathbf{z}_{\mathcal{M}})$; $(\boldsymbol{\tau}, \boldsymbol{\phi}) \leftarrow \mathcal{D}(\mathbf{x}; \mathbf{L}_{\mathrm{mask}})$
  $E \leftarrow \mathcal{G}(\boldsymbol{\tau}, \boldsymbol{\phi}; \mathbf{b}_0)$ ▷ joint arm+base LM solve, Eq. (3)
  $\mathbf{x} \leftarrow \mathrm{ODEStep}\big(\mathbf{x},\ \mathbf{v} - w\,\nabla_{\mathbf{x}} E \odot \mathbf{m}_{\mathrm{hand}},\ t\big)$
end for
$\mathbf{L}[\mathcal{M}] \leftarrow \mathbf{x}$
if $s \geq s_{\mathrm{start}}$ **then** decode $\mathbf{L}$; $\mathbf{b}_0 \leftarrow \mathrm{mean}_t(\mathbf{b}^{t*})$
end if

C. Retargeting and Policy Learning

Retargeting. Unless stated otherwise, the target is a Unitree G1 with two XHands: 15 controllable upper-body DoFs (torso and both arms), fixed lower body, and 12 DoFs per hand. PyRoki [24] solves the guided wrist trajectories to arm joint trajectories from the optimized base; dex-retargeting [26] maps MANO fingers to XHand joints; and contact retargeting following DexMachina [29] yields the reference contacts $\hat{\mathbf{c}}_t$. Generated object motion supplies the reference object pose and articulation used in observations, rewards, and terminations.

Policy. We train an MLP policy (four layers, 512–512–256–128, ELU) with PPO [38] (rl-games) in Genesis with 8192 parallel environments for 2000 iterations on one RTX 4090 (≈4 h). Observations comprise four groups: short histories of joint positions, velocities, and actions; the episode phase; sparse future reference joint offsets $\hat{\mathbf{q}}^{\mathrm{robot}}_{t+\Delta} - \mathbf{q}^{\mathrm{robot}}_t$ ($\Delta \in \{1, 2, 4, 8, 16\}$); and, for the object, its part poses, articulation, and future tracking errors against the generated trajectory. Actions are 39-D: 15-D *delta* targets for torso and arms and 24-D *absolute* finger targets. We never actuate the object, so unlike DexMachina we use no virtual-object-controller curriculum, and no reward or entropy curriculum.

Decoupled arm and hand rewards and terminations. We split imitation into an upper-body stream $r_{\mathrm{imi,upper}}$ over 15 DoFs and a dexterous stream $r_{\mathrm{imi,dex}}$ over 24 DoFs. Coarse free-space reaching and contact-rich finger control differ in error scale, so a single term over all 39 DoFs biases optimization toward whichever subsystem dominates the error. Each stream is a joint-tracking error against the retargeted reference with its own weight and temperature (w_{bc}=1.0, β=30; w_{finger}=0.15, β=20). The total reward adds an object-tracking task term r_{task} (w=1.3) on position, orientation, and articulation, a contact term (w=0.7) matching retargeted contact locations, and standard action, rate, velocity, acceleration, and torque penalties. Termination

is decoupled in the same way, on the half-squared joint error $\frac{1}{2}\Delta q^2$: the upper-body stream terminates on its *maximum* over joints (ϵ=0.12, a 0.49 rad deviation), as a large arm failure ends the rollout, and the hand stream on its *mean* over finger joints (ϵ=0.2), since an isolated finger error is not catastrophic. Episodes also end if the object falls below the table or its half-squared articulation error exceeds ϵ_a=0.4.

[NOTE:] Implementation Details subsection — recommended, and now free since arXiv has no page limit. The skill puts hyperparameters at the end of Method or in a dedicated subsection; right now they are scattered across three places (IK weights in Sec. III-B, reward weights and temperatures in Sec. III-C, PPO settings in the **Policy.** paragraph), which makes both subsections read as specification rather than argument, and still leaves three values a reader needs unstated. **Uncomment the block below** and delete the inline hyperparameter lists from Secs. III-B and III-C — their argument sentences ("fixed once for both embodiments and all tasks", "coarse reaching and finger contact differ in error scale") then stand alone, which is what they deserve.

Three values are genuinely missing from the paper and are marked `[CONFIRM]` below: *(i)* the guidance weight w in Eq. (1) — it appears in the equation and in Alg. 1 and is never given a number anywhere, in this draft or the CoRL one, which is a reproducibility hole a reviewer can legitimately call out; *(ii)* s_{start}, the outer step at which base-prior refresh begins; *(iii)* the number of base-prior updates *inside* the sampling loop (the protocol gives two for the standalone evaluator; the loop is unstated). Take all numeric values from the current code, not from either manuscript.

IV. Experiments

We answer four questions. (Q1) Does guidance *during* sampling earn its cost relative to post-hoc IK refinement and compute-matched best-of-N selection? (Sec. IV-A) (Q2) How does guidance change kinematic conditioning and generation quality? (Sec. IV-B) (Q3) How much of the downstream gain comes from each component? (Sec. IV-C) (Q4) Does the same objective transfer across embodiments and generators? (Sec. IV-D)

Protocol. Downstream results use two protocols on the G1 with XHands: a fixed four-task suite (box open, waffle-iron lift, mixer open+close, microwave lift+place; Tables I and III) and 50 randomly sampled tasks with no pre-selection. Of the latter, the 10 whose unguided reference falls below $\sigma_{\min}$<0.1, and consequently carries high IK energy, form a hard subset (Sec. IV-C). Retargeting, rewards, and PPO follow Sec. III-C in every condition; only the reference differs. Q1 uses 51 seed-matched test prompts that include the four-task suite, Q2 400 ARCTIC test prompts, Q3 the four-task suite and the 50 random tasks, and Q4 the prompt sets stated in Sec. IV-D. We recompute every IK metric for every condition with one standalone instance of the IK solver of Sec. III-B (optimized base, two base-prior updates from the same initial prior), so the rows of a table are directly comparable.

Success requires the object's position, orientation, and articulation to stay within 2 cm, 40°, and 10° of the reference at every step of the episode. Success rates (SR) are means over 5 training seeds, each evaluated on 10 episodes of B parallel environments with randomized object initialization per task ($10B$ rollouts per seed and task), with the standard error over seeds in Tables I and III. Kinematic metrics are the IK energy E (Eq. 3) and the worst-case Jacobian conditioning over the sequence (minimum singular value $\sigma_{\min}$, maximum condition number) on the target robot's upper-body chain.

Generation quality uses the standard text-to-motion metrics (FID, KID, diversity, top-3 retrieval accuracy, matching score) and geometric metrics (contact ratio, penetration depth, jitter).

TABLE I: Alternatives to IK guidance on 51 prompts (medians). Post-hoc refinement and best-of-N selection reach lower IK energy than guidance but train worse policies; best-of-32 matches the guidance budget. Cost: sampling time per reference in native samples (t_1=5 s); Δ_{wrist}: wrist displacement (cm) from the seed-matched unguided sample; Pen.: penetration loss; SR: success rate on the four-task suite, mean ± s.e.m. over 5 seeds.

Method	Cost	$E\downarrow$	Δ_{wrist}	Pen.↓	SR %↑
Unguided, fixed base	t_1	2.30	–	0.52	68.1 ± 0.3
+ base optimization	t_1	1.89	0	0.52	73.3 ± 0.5
+ IK guidance (ours)	$32\,t_1$	1.81	**0.3**	0.53	**79.7** ± **0.4**
Post-hoc IK	t_1+IK	**0.35**	32.6	0.59	32.8 ± 0.3
Best-of-8 by E	$8\,t_1$	1.36	13.2	0.52	73.1 ± 0.6
Best-of-32 by E	$32\,t_1$	1.21	13.7	0.51	74.0 ± 0.6

A. Is Guidance During Generation Necessary? (Q1)

Guidance costs 32× a native sample, so each alternative in Table I receives the same budget, the same 51 prompts, and the same IK solver and base optimizer. **Base-placement** optimization alone keeps the sample and re-solves only the base. This cost is paid once per reference and amortized over ≈4 h of policy training it supports, so the wall-clock comparison is a worst case rather than a deployment constraint. **Post-hoc IK** refinement optimizes the wrist channels of a finished unguided sample against the same energy (Adam, 100 steps). **Best-of-N** draws N unguided samples with fixed seeds and keeps the one with the lowest terminal E under the same optimized base.

Base placement is the cheapest large gain. Solving for the base with a single unguided sample reduces the median energy from 2.30 to 1.89 and raises the success rate from 68.1 to 73.3 at no sampling cost (Fig. 3). Guidance adds a smaller, consistent correction on top: it lowers the energy of the seed-matched sample on 40 of the 51 prompts (78%; median 1.81) and raises the success rate to 79.7.

IK energy is not the quantity to optimize. Post-hoc refinement reaches the lowest energy of any method (E=0.35) but moves the wrists by 33 cm after finger and object motion are fixed, raising penetration from 0.52 to 0.59 and collapsing the success rate to 32.8. Selection lowers the energy further than guidance at every budget: 1.36 at N=8 and 1.21 at the compute-matched N=32, and wins on 88–94% of prompts whether the prompt is easy, medium or hard (see video).

[FIX] "(see video)" cannot support a stratified percentage — a video shows motions, not a per-difficulty win rate, so this reads as a citation to evidence that does not exist. The range 88–94% is itself the stratification (three bins), so stating the bins turns an unsupported claim into a supported one at no cost. Use the first version if you have the three numbers to hand; the second if you would rather not open the question at all.

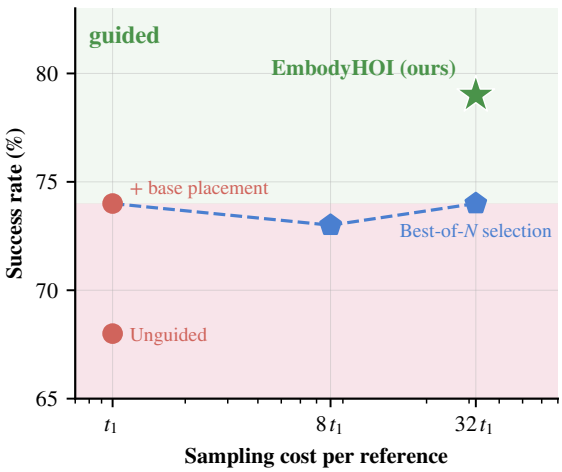


Fig. 3: Success rate against sampling cost per reference on the four-task suite. Base placement lifts the unguided baseline at no cost, best-of-N selection does not measurably improve on it at 32 times the cost (0.7 ± 0.8 points), and IK guidance is the only method that clears the best baseline.

REPLACE: *and wins on 88–94% of prompts whether the prompt is easy, medium or hard (see video).*
WITH: and wins on 88--94% of prompts --- 88% on the easiest third by unguided E, 94% on the hardest --- so its advantage does not shrink where reachability is worst. *(substitute your actual per-bin numbers)*; or, if the stratification is not to hand, and wins on 88--94% of prompts across the difficulty range.

The reason is variance. Energy varies across samples of one prompt (s.d. 0.36) far more than guidance shifts it (median per-prompt shift 0.04), so drawing more samples finds a low-energy trajectory more easily than correcting one sample does. Yet the policies trained from selected references succeed less often than from guided ones (73.1 at N=8, 74.0 at N=32, vs. 79.7), a gap of 5.7 ± 0.7 points at matched compute. The energy measures wrist reachability only; it is blind to whether fingers and object still form a coherent interaction around the wrists it prefers.

Guidance corrects, selection re-draws. The selected best-of-N sample lies as far from the seed-matched one as any other sample (13 cm), whereas guidance moves the wrists of the seed-matched sample by 0.3 cm and leaves penetration and contact unchanged. Only guidance edits the sample already in hand, which is what matters when that sample is one a user chose. Base-placement optimization is therefore the baseline to report alongside guidance: it is free, selection is a no-op on top of it at either budget (-0.2 ± 0.8 at N=8, 0.7 ± 0.8 at N=32), and raising N from 8 to 32 does not help either (0.9 ± 0.8). Energy is diagnostic only at fixed interaction content, which is why it is meaningful for a seed-matched guided-vs- unguided pair and meaningless across independently drawn samples.

[FIX] Typo (stray space inside the hyphenated compound), and the sentence is currently the last line of the paragraph when it is the resolution of the whole subsection — it belongs at the front of the "IK energy is not the quantity to optimize" paragraph, where it tells the reader in advance why the coming numbers do not contradict your abstract.
REPLACE: *Energy is diagnostic only at fixed interaction content, which is why it is meaningful for a seed-matched guided-vs- unguided pair and meaningless across independently drawn samples.*
WITH: Energy is diagnostic only at fixed interaction content: it is meaningful for a seed-matched guided-vs-unguided pair, and meaningless across independently drawn samples.
— and move it to immediately after the bold heading **IK energy is not the quantity to optimize.**, before "Post-hoc refinement reaches the lowest energy…".

B. Kinematic Conditioning and Generation Quality (Q2)

Conditioning. Base placement and guidance remove different failure modes. On 400 ARCTIC test prompts, optimizing the base alone cuts the share of near-singular unguided samples ($\sigma_{\min} < 0.03$) from 2.6% to 0.2% and the median IK energy from 2.59 to 2.19 (Fig. 5); after that, $\sigma_{\min}$ ($\approx$0.086) and the condition number ($\approx$26) are as good as the chain allows, and guidance leaves them there. Guidance instead removes the residual error that survives retargeting: it lowers the energy on 83% of prompts, by 2.9% on average and by 8% on the hardest quartile (Fig. 4), and the G1's wrist tracking error after retargeting falls from 8.9 cm and 20° to 0.6 cm and 4° (Table II). On the remaining 17% of prompts the energy rises by 1.2% at the median with the wrists within 0.5 cm of the unguided sample; on 2 of the 51 seed-matched prompts of Sec. IV-A the guidance step propagated through later unmasking rounds into a different trajectory. No sample is filtered, so the success rates in Secs. IV-A and IV-C include these cases. We do not measure a per-prompt relation between energy and success.

Generation quality. Guidance does not degrade the prior (Table II). On the identical 400-prompt subset, guided samples have lower FID than unguided (0.545 vs. 0.667), better top-3 retrieval (0.872 vs. 0.852), unchanged contact ratio (83.1 vs. 83.5%) and diversity (19.85 vs. 19.77), lower penetration (11.5 vs. 12.2%), and marginally higher KID (0.003 vs. 0.001). Guidance necessarily integrates the flow ODE itself (20 RK4 steps, Alg. 1), so this compares our sampler against MAD-HOI's released one rather than ablating the IK term alone, and the 0.12 FID gain belongs to the guided sampler as a whole. The IK objective reshapes wrist trajectories toward reachable configurations without collapsing the semantic content of the interaction (see supplementary video).

Sensitivity to weights. The weights are set once and frozen across all tasks and both robots (Sec. IV-D). Zeroing the rest-pose prior, zeroing or tripling the manipulability term, or scaling the root prior by 0.3 on the same 400 prompts moves the median IK energy by at most 0.03 and $\sigma_{\min}$ by less than 1% (Fig. 5), and every setting degrades FID (0.615–0.820 vs. 0.545) and top-3 retrieval (0.835–0.869 vs. 0.872), the manipulability term is the most sensitive of the four: zeroing λ_{manip} accounts for the 0.820 end of the FID range. We did not retrain policies under perturbed weights, so downstream sensitivity is bounded only by Table III, where dropping a whole weight group costs 2.9 to 4.0 points.

[NOTE:] Two further things in this paragraph, both cheap. *(a)* Say that the defaults were fixed *before* the sweep rather than chosen by it — otherwise a reviewer reads eight hand-set weights whose default happens to win every column as retrofitted. *(b)* The sweep perturbs four weights out

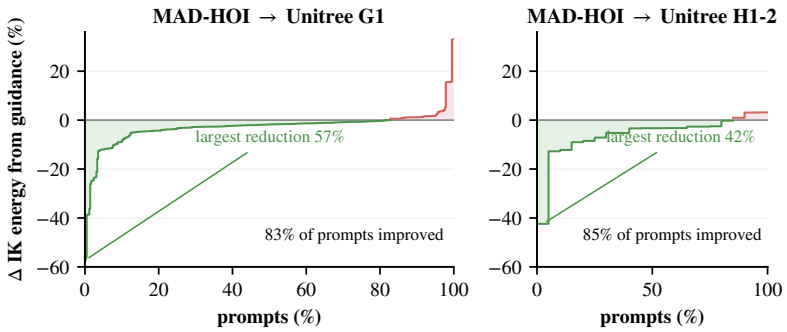


Fig. 4: Change in IK energy that guidance applies to each seed-matched sample. Guidance makes 83–97% of samples more reachable depending on the robot–generator pair; the remaining increases are small (median 1%), with two outliers on the G1 where sampling propagated the correction into a different trajectory (Sec. IV-B).

TABLE II: Guidance degrades no generation metric while removing most of the retargeting error. Left: generation quality of unguided and IK-guided MAD-HOI on the *same* 400 ARCTIC test prompts with one evaluator. Right: wrist tracking error of the retargeted G1 after PyRoki IK on the randomly sampled tasks.

	FID↓	KID↓	Div.↑	Top-3↑	Match↓	Contact %↑	Pen. <1 cm %↓	Jitter↓	Wrist err. (cm)↓	Wrist err. (°)↓
Unguided	0.667	0.001	19.77	0.852	3.99	83.5	12.2	0.287	8.9	20
IK-guided	**0.545**	0.003	**19.85**	**0.872**	**3.97**	83.1	**11.5**	0.293	**0.6**	**4**

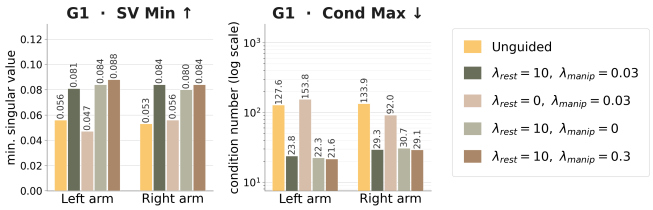


Fig. 5: Base placement, not the IK weights, removes near-singular configurations. Worst-case conditioning of the G1 per arm over the ARCTIC prompts: minimum Jacobian singular value (↑) and maximum condition number (↓, log scale). Unguided samples sit at the fixed base, guided samples at the optimized base under four IK-weight settings, which overlap.

of eight; saying why those four (the other four are task residuals with no free scale, presumably) forecloses "why not sweep w_{pos}?".

C. Downstream Policy Learning and Ablations (Q3)

Randomly sampled tasks. Guided references raise the average success rate on 50 randomly sampled tasks from 68% to 79%, and on the 10 worst-conditioned of them from 57% to 82%.

[FIX] The 25-point hard-subset gain is your single most persuasive number and it is the only headline result with no error bar. It is also the most attackable: 10 tasks selected *post hoc* by a criterion ($\sigma_{\min}$<0.1) computed on the unguided reference, which is exactly the pattern reviewers associate with subset-shopping — even though here the criterion is stated in advance, principled, and precisely the population your method targets. Two clauses neutralize it permanently: give the s.e.m. (you have 5 seeds) and say the criterion is fixed before training.

REPLACE: *and on the 10 worst-conditioned of them from 57% to 82%.*

WITH: `and on the 10 worst-conditioned of them --- selected by $\svmin$ of the` **`unguided`** `reference before any policy is trained --- from 57.x $\pm$ y to 82.x $\pm$ y.` *(fill in the s.e.m. over your 5 seeds)*

The 50 tasks are drawn with no selection across all 11 ARCTIC objects, and we train policies from a guided and an unguided reference of each (5 seeds). The gain concentrates where reachability is poor: most randomly drawn prompts are already reachable, and the aggregate gain comes from the poorly conditioned minority (Sec. IV-A).

TABLE III: Reference and RL-recipe ablation on the G1 four-task suite (box open, waffle-iron lift, mixer open+close, microwave lift+place); mean ± s.e.m. over 5 seeds, 10 episodes per task. Rows A–D vary the reference source, the base placement, and the RL recipe (DexMachina-style [29] vs. ours); the lower half ablates our components from D.

	Reference → retarget / RL	Avg. SR %↑
A	ARCTIC GT, fixed base → DexMachina-style	66.8 ± 0.3
B	Unguided, fixed base → DexMachina-style	68.2 ± 0.9
C	Unguided, optimized base → ours	73.6 ± 0.9
D	Guided (both groups) → ours (full)	**79.3 ± 0.6**
	D, base-placement group only	76.4 ± 0.3
	D, arm-conditioning group only	75.3 ± 0.3
	D, shared imitation reward	74.1 ± 0.6
	D, shared reward, no contact retarg.	73.0 ± 0.4

Controlled comparison and ablation. Two changes move downstream success on the four-task suite, and they are comparable in size (Table III). A→B holds reachability fixed and varies only the reference source: swapping the ARCTIC demonstration for an unguided generated sample changes the success rate by 1.4 points, so generation supplies diversity but not reachability. B→C then introduces our RL recipe together with base placement and adds 5.4 points, and C→D adds IK guidance on top of it for a further 5.7. Neither half alone accounts for the method's gain. The "DexMachina-style" rows re-implement their object-side recipe on our 15-DoF arm chain; they are not a run of their released floating-wrist system. Because that system commands a free-floating wrist with no arm chain, it never faces the reachability

problem and is not a like-for-like baseline: its 86.1% on the same tasks is an upper bound obtained by assuming away the constraint we study, and the gap to our 79.3% is that constraint's cost rather than a deficiency of the RL recipe.

The two weight groups of the IK objective contribute comparably to the 5.7 points that C→D adds. Zeroing one group at a time, dropping base placement costs 4.0 points (79.3 to 75.3) and dropping the four arm-conditioning terms costs 2.9 (79.3 to 76.4); read the other way, base placement alone adds 2.8 over row C and arm conditioning alone 1.7, so the two together achieve more than the sum of their parts. The comparison against row C is the one that matters for the design: row C applies the same base optimization once, after generation, so re-solving the base inside the sampling loop rather than after it (Sec. III-B) accounts for 2.8 of those points. Base placement is therefore not preprocessing that any pipeline could bolt on, but a term in the guidance objective whose value comes from being inside the loop. A shared imitation reward costs 5.2 points, and dropping contact retargeting only costs 1.1 more.

Reward design. Averaged over the suite, imitation alone reaches 44.5%, the object-tracking term alone 58.2%, a single imitation term over all 39 DoFs 74.1%, and our decoupled arm and hand streams 79.3%. Decoupling wins on every task, with the largest margins where coarse reaching and finger contact differ most in error scale (supplementary video).

Feasibility beyond kinematics. Guidance does not reason about dynamics, but reachable references are also closer to executable ones: on 50 of the 51 unique prompts of Sec. IV-A, 72% of guided references stay within the G1's joint velocity limits versus 68% of the seed-matched unguided references (median peak-velocity ratio 0.20 vs. 0.24).

[FIX] Two problems in one sentence. (i) "50 of the 51" silently drops a prompt — an unexplained exclusion in the one paragraph claiming a bonus property reads as selective reporting even when the cause is mundane. (ii) 72% vs. 68% of 50 prompts is 36 vs. 34 sequences, a two-sequence difference with no interval; the skill's review rule §3.1 asks whether gains are meaningful rather than statistically tiny, and this one is not distinguishable from noise. The median peak-velocity ratio uses every sequence and is the stronger statistic, so lead with it and drop the thresholded percentages.

REPLACE: *on 50 of the 51 unique prompts of Sec. IV-A, 72% of guided references stay within the G1's joint velocity limits versus 68% of the seed-matched unguided references (median peak-velocity ratio 0.20 vs. 0.24).*

WITH:
```
across the prompts of Sec. IV-A, guided references have a lower median peak-velocity ratio against the G1's joint limits than their seed-matched unguided counterparts (0.20 vs. 0.24).
```
If you keep the percentages, state why one prompt is excluded, e.g. `(one prompt is excluded because its unguided IK solve did not converge).`

D. Transfer Across Embodiments and Generators (Q4)

Because guidance acts at inference on a frozen generator, changing the robot changes only the IK module. On the Unitree H1-2, whose arms are 68.5 cm long compared to G1's 45 cm, guidance lowers the IK energy of the seed-matched sample on 46 of 50 prompts (median 2.06 → 1.95, mean −6%) while moving wrists by 0.3 cm. Both conditions stay far from singularity ($\sigma_{\min} \approx 0.13$) because H1-2's longer arms are closer to human proportions. The base moves in the opposite direction to G1's (Fig. 6 and supplementary video). The gain carries through to policy learning: on the same four-task suite, guidance raises H1-2 success rate from $62.3 \pm 0.4\%$ to $75.7 \pm 0.5\%$ (5 seeds, 10 episodes per task), a 13.4% gain comparable to the G1's 11.6% gain (68.1 to 79.7, Table I) despite the different arm geometry.

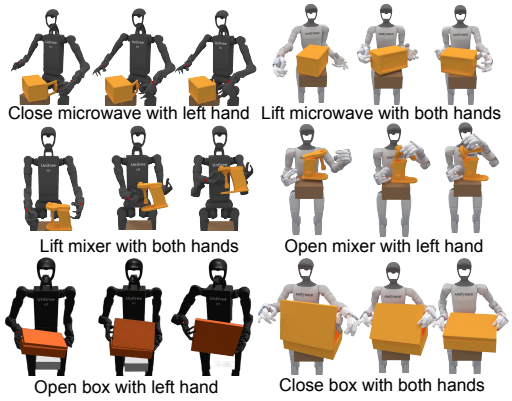


Fig. 6: One generator and one IK objective place two humanoids differently: from the same initial placement, guidance moves the H1-2 base (left; 178 cm, 68.5 cm arms) away from the object to avoid over-extension and the G1 base (right; 132 cm, 45 cm arms) closer and higher to stay reachable.

[FIX] Percentage points, not percent. 62.3% to 75.7% is a 13.4-*point* gain (and a 21% relative gain). Robotics reviewers read this closely because the two readings differ by a factor of 1.6, and writing "%" where you mean "points" looks like the error is in your favour.

REPLACE: *a 13.4% gain comparable to the G1's 11.6% gain (68.1 to 79.7, Table I) despite the different arm geometry.*

WITH:
```
a 13.4-point gain comparable to the G1's 11.6 points (68.1 to 79.7, Table \ref{tab:necessity}) despite the different arm geometry.
```

[NOTE:] *The strongest sentence in this subsection is the shortest one.* "The base moves in the opposite direction to G1's" is your best evidence that the module reasons about embodiment rather than applying a learned offset — one objective, unchanged, placing a tall robot back and a short robot forward. It currently sits between two energy statistics where it reads as an aside. Consider:
```
Notably, the base moves in the opposite direction to G1's (Fig. \ref{fig:cross}): the longer arms over-extend and must stand back, where the G1's must close in. No term in the objective encodes either behaviour; both fall out of the same residual.
```

The module is also generator-agnostic. Text2HOI [22] represents the wrist pose identically to MAD-HOI, so the IK objective and the base prior transfer without modification. Applied to its DDPM sampler as a gradient step on prediction x_0, the guidance lowers the IK energy on 48 of 50 ARCTIC prompts (median 2.36 → 2.34, mean −1.3%) with at most 0.4 cm of wrist adjustment. The same objective on the H1-2 lowers Text2HOI's energy on all 50 ARCTIC prompts (median 1.77 → 1.75, mean −1.2%), again with the wrists within 0.5 cm. The effect is smaller than on MAD-HOI because Text2HOI's contact refiner post-processes the hands, but its direction is consistent across generators and robots. We therefore do not depend on MAD-HOI: its checkpoint is public [10], we will release the guidance code, all generated

references, and the trained policies.

V. Limitations and Conclusion

EMBODYHOI makes a frozen HOI generator produce motions a target humanoid can reach, by running a differentiable IK solve inside its sampling loop and optimizing the robot's base. Policies trained on the guided references succeed more often than on unguided ones (79% vs. 68% on 50 random tasks, 76% vs. 62% on the H1-2), with base placement and wrist conditioning contributing about equally.

[FIX] Two issues. (i) *Terminology:* this is the only place the component is called "wrist conditioning"; Table III calls it "arm-conditioning group" and contribution (2) calls it "base placement … in the same objective". Pick one name and use it everywhere — *arm conditioning* matches the table. (ii) *"About equally" is not what your numbers say:* 4.0 vs. 2.9 points with $\pm$0.3 s.e.m. are distinguishable, and the truthful version is more interesting, because the larger contributor is also the free one.

REPLACE: *with base placement and wrist conditioning contributing about equally.*

WITH: `with base placement the larger of the two contributors (4.0 vs. 2.9 points) and also the cheaper, since it adds no sampling cost.`

[NOTE:] *The conclusion still restates what the method does and never says what the paper taught.* The skill's conclusion template asks for a practical-impact or insight element, and you have an unusually good one that currently lives only in Sec. IV-A: it generalizes past EMBODYHOI to anyone building a retargeting or reference-generation pipeline. Add as the last sentence of the first paragraph: `More broadly, our comparison against post-hoc refinement and best-of-N selection is a caution for reference-generation pipelines in general: a kinematic feasibility metric is a valid diagnostic only when the rest of the interaction is held fixed, and driving it down directly buys feasibility at the cost of the interaction it was meant to preserve.` Guidance is kinematic: better reachability does not imply dynamic or contact feasibility, and the RL stage absorbs what guidance does not model. The cost is a 32$\times$ slower sampler, paid once per reference; restricting guidance to a subset of the refinement steps would reduce it.

Our scope is limited in two ways. The lower body is fixed: we optimize where the robot stands, and not how it gets there. Integrating a locomotion controller [4], [39] is left for future work. All evaluation is performed in simulation on ARCTIC's articulated objects; real-world rollouts are the immediate next step. Beyond scope, fine-tuning the generator on its own guided samples would distill the guidance into the prior and remove the inference cost, at the price of one generator per embodiment.

[NOTE:] arXiv appendix plan. Below is a paste-ready skeleton. Sources are your CoRL draft at `CoRL_2026/corl_2026_template_submission/` — `example.tex` lines 858–1123 and the whole of `appendix_tables.tex`, which can be copied across almost unchanged (it is three tables plus a hyperparameter table and needs only a `\input{appendix_tables}`). Take every number from the current code, not from either manuscript.

Priority order, by what a reviewer or re-implementer actually needs: **(1)** The three RL specification tables (observations, rewards, terminations) — `appendix_tables.tex` already has them, and the termination table is the only place the half-squared-error convention behind ϵ=0.12 is written as a formula rather than asserted. **(2)** PPO hyperparameters. **(3)** Action parameterization: the joint delta scale, PLAI α, and action delay are absent from the current paper entirely and are exactly the details that decide whether someone reproduces your numbers. **(4)** The qualitative figures you cut — `guided-vs-unguided.pdf` is the one that shows the semantic content survives guidance, which the main text currently asserts and defers to the video; a static figure is what a reviewer looks at. **(5)** The H1-2 singularity grid.

Content the main text now promises but nothing backs — each is a table the appendix should carry: the 86.1% DexMachina floating-wrist number (Sec. IV-C, cited in prose with no table); the stratified best-of-N comparison behind "88–94% … easy, medium or hard"; the per-setting weight-sweep numbers behind the FID range 0.615–0.820; the per-task breakdown of the 50 random tasks and the 10-task hard subset; and the paired bootstrap CI for the 79 vs. 73 gap. Of these, the stratified table is the one your Sec. IV-A argument rests on.

One thing to drop rather than port: the CoRL appendix's "End-to-End Success of Guided vs. Unguided" section states outright that it "intentionally choose[s] tasks which show the highest improvement in singularity metrics". The ICRA draft replaced that with the unselected 50-task protocol, which is strictly stronger. Porting the old section would reintroduce a selection bias you have already designed out.

Acknowledgments

LLMs were used for copy-editing and debugging scripts; all technical content, experiments, and analyses are the authors' own.

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